HO CHI MINH CITY UNIVERSITY OF TRANSPORT (UTH)
FACULTY OF INFORMATION TECHNOLOGY

Capstone Project Document

Horse Racing Tournament Management System

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Supervisor	Lecturer Nguyen Van Chien
Capstone Project code	Hourse Racing Tournament

- Ho Chi Minh City, July 2026 -

Contents

Acknowledgement

We would like to express our sincere gratitude to the faculty of Ho Chi Minh City University of Transport (UTH), and especially to our supervisor, Nguyen Van Chien, for their dedicated guidance and support throughout the development of this Capstone Project.

We would also like to thank every member of our team for their effort and close collaboration, which made it possible to complete the "Horse Racing Tournament Management System" project on schedule.

Definition and Acronyms

Term / Abbreviation	**Definition**
MVP	Minimum Viable Product
SRS	Software Requirement Specification
SDD	Software Design Description
ERD	Entity Relationship Diagram
UC	Use Case
BR	Business Rule
CR	Common Requirement
MSG	Application Message
UI	User Interface
API	Application Programming Interface
DB	Database

I. Project Introduction

1.1. Overview

1.1.1 Project Information

- **Project name:** Horse Racing Tournament Management System
- **Project code:**
- **Group name:** Group 3
- **Software Type:** Web application

1.1.2 Project Team

Full Name	Role	Email	Mobile
Nguyen Van Chien	Supervisor	chiennv@ut.edu.vn	
Ha Duy Hung	Leader	haduydung0912@gmail.com	
Dinh Thuy Anh	Member	anhdt1497@ut.edu.vn	
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Nguyen Gia Huy	Member	nhuy22174@gmail.com	
Doan Thi Thu May	Member	maydtt6742@ut.edu.vn	
Bui Le Quang Huy	Member	huyblq0064@ut.edu.vn	

Table 1.1: Project Team

1.2. Product Background

In the context of Vietnam's growing interest in sports entertainment and digital transformation, horse racing organizations are increasingly facing the need to manage tournaments, participants, horses, jockeys, and race registration processes more efficiently. However, many racing-related activities may still depend on manual management methods such as paper forms, scattered Excel files, direct communication, or disconnected systems. This can lead to difficulties in managing tournament information, horse and jockey profiles, owner registrations, approval status, and communication between horse owners, jockeys, referees, and administrators.

Recognizing these limitations, a student team has proposed the development of a Horse Racing Management System. This system aims to digitize the main workflow of horse racing event management, from tournament creation and participant registration to approval, race organization, and result tracking. The target users include horse owners who register horses for tournaments, jockeys who participate in races, administrators who manage tournaments and registration requests, referees who monitor race results, and viewers or related users who need to follow tournament information in a transparent and convenient way.

1.3. Existing Systems

1.3.1 Prism Horse

Link: `https://prism.horse`

Description: A horse racing management platform supporting race events, horse information, and participant management.

Actors:

- Horse Owners
- Jockeys
- Administrators
- Referees

Features:

- Horse and jockey management.
- Tournament organization.
- Race registration.
- Participant management.

Pros:

- Designed specifically for horse racing activities.
- Supports race and participant management.

Cons:

- Limited owner–jockey collaboration workflow.
- No invitation management mechanism.
- Limited registration status tracking.
- Lack of integrated communication features.

1.3.2 RaceDB

Link: `https://www.racedb.com`

Description: A race event management system that supports competition registration, participant management, race scheduling, and result tracking. The platform is primarily designed for cycling and sporting competitions.

Actors:

- Participants
- Event Organizers
- Administrators
- Race Officials

Features:

- Online participant registration.
- Race scheduling and event management.
- Result recording and ranking.
- Participant and competition tracking.

Pros:

- Supports online registration and race management.
- Provides race scheduling and result tracking.
- Offers a centralized system for competition administration.

Cons:

- Not specifically designed for horse racing.
- Does not support horse owner, horse, and jockey relationships.
- Lacks invitation and approval workflows between horse owners and jockeys.
- Does not provide horse profile management and ownership information.
- Limited support for horse racing officials and tournament-specific workflows.

1.4. Business Opportunity

The horse racing industry is gradually embracing digital transformation as racing organizations and clubs seek more efficient ways to manage tournaments, participants, and race operations. However, many horse racing activities still rely on manual registration processes, paper documents, and fragmented communication between horse owners, jockeys, and organizers. This creates an opportunity to develop a specialized management system that can streamline tournament registration, improve participant coordination, and enhance transparency throughout the racing process.

The proposed Horse Racing Management System aims to optimize race organization by supporting online tournament registration, horse and jockey management, invitation workflows, and registration approval processes. The system can reduce administrative workload, minimize registration errors, and improve communication among stakeholders.

The system will compete in the sports event management and competition management software market while specifically targeting the horse racing domain, where existing solutions often lack comprehensive support for horse owner–jockey collaboration, race registration workflows, and tournament management requirements. By focusing on the unique characteristics of horse racing activities, the system has the potential to provide a more specialized and efficient solution for racing organizations and participants.

1.5. Software Product Vision

For horse racing organizations seeking to digitalize tournament operations, the Horse Racing Management System is a web-based platform that enables horse owners to register horses, invite jockeys, and participate in racing tournaments through an integrated online process. Jockeys can manage invitations and confirm participation, while administrators and referees can efficiently monitor registrations, approve requests, and oversee race activities.

Unlike traditional management methods that rely on paper documents, manual registration, and fragmented communication, the Horse Racing Management System reduces administrative workload, improves the accuracy and transparency of tournament management, and enhances collaboration among horse owners, jockeys, referees, and administrators. By digitalizing registration workflows, participant information, and competition management, the system also establishes a foundation for future services such as performance analytics, race statistics, online event management, and data-driven decision making in horse racing organizations.

1.6. Project Scope & Limitations

1.6.1 Major Features

Major Features for Admin

FE-01: Login/Logout

FE-02: Manage user accounts

Major Features for Horse Owner

FE-01: Register horse for tournament
FE-02: View registration status

1.6.2 Limitations & Exclusions

LI-1: Write limitation here.

II. Software Project Management Plan

This chapter describes project planning, scheduling, risk management, communication and configuration management.

III. Software Requirement Specification

3.1. Functional Requirements

3.1.1 Horse Owner Registration Screen Flow

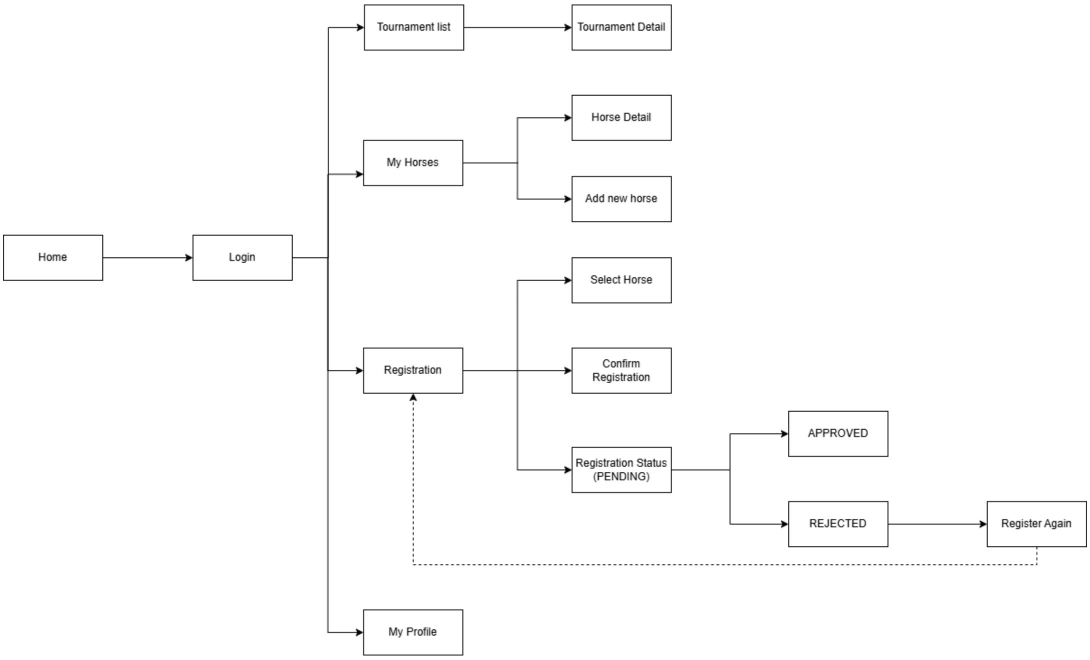

Figure 3.1: Horse Owner registration screen flow

3.1.2 Screen description

#	Feature	Screen	Description
1	General	Home	Landing page for owners; overview and navigation to login/register.
2	General	Login	Authentication screen allowing the owner to access the system with secure credentials.
3	Dashboard	Owner Dashboard	The central hub displaying the owner's horses, upcoming races, and quick actions.
4	Registration	Registration Form	Form to register as a horse owner: personal details, farm details, contact and horse information.
5	Registration	Confirmation	Review and confirm entered data before submission.
6	Registration	Success	Acknowledgement that registration was received and next steps.

Table 3.1: Horse Owner screen descriptions

3.2. Web Application

3.2.1 Horse Owner

Login

Role: Horse Owner

- **Function trigger:** The owner enters account credentials to log in.
- **Function description:**
  - **Purpose:**
    1. Verify the owner's identity.
    2. Grant access to owner functionalities.
  - **Data processing:**
    1. The owner enters username and password.
    2. The system validates credentials.
    3. The system identifies the OWNER role.
    4. The system creates a login session.
- **Function details:**
  - Successful login redirects to Owner Dashboard.
  - Invalid credentials display an error message.

  - Users without accounts may register.

Screen layout:

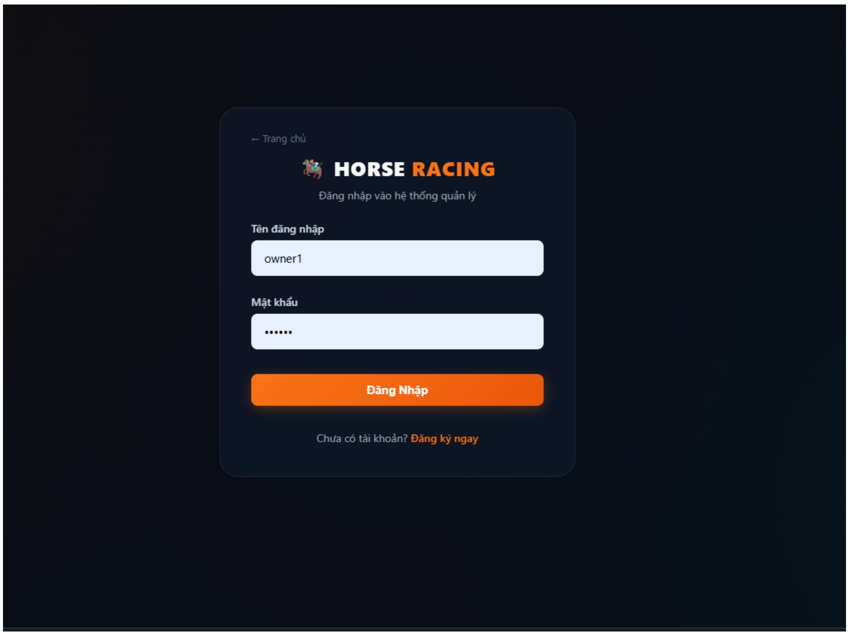

Figure 3.2: Horse Owner Login

3.2.2 Manage Horses

Role: Horse Owner

- **Function trigger:** Select "Quan ly Ngua".
- **Function description:**
  - **Purpose:**
    1. Register new horses.
    2. Manage owned horses.
  - **Data processing:**
    1. Enter horse name.
    2. Enter age.
    3. Select breed.
    4. Select gender.
    5. Save horse information.
- **Function details:**
  - Add new horses.
  - Edit horse information.
  - Delete horse information.
  - Display horse list.

Screen layout:

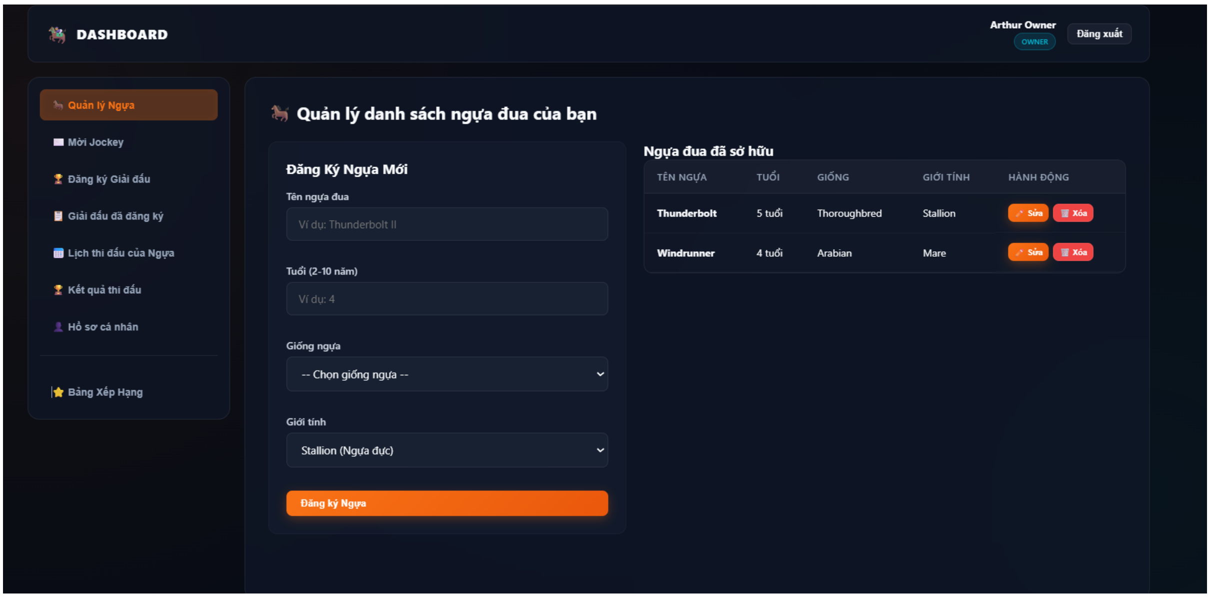

Figure 3.3: Manage Horses

3.2.3 Invite Jockey

Role: Horse Owner

- **Function trigger:** Select "Moi Jockey".
- **Function description:**
  - **Purpose:**
    1. Invite jockeys.
    2. Create horse-jockey partnerships.
  - **Data processing:**
    1. Select jockey.
    2. Select horse.
    3. Select tournament.
    4. Enter invitation message.
    5. Send invitation.
- **Function details:**
  - Invitation status is displayed.
  - Pending invitations can be monitored.
  - Accepted invitations may be used for registration.

Screen layout:

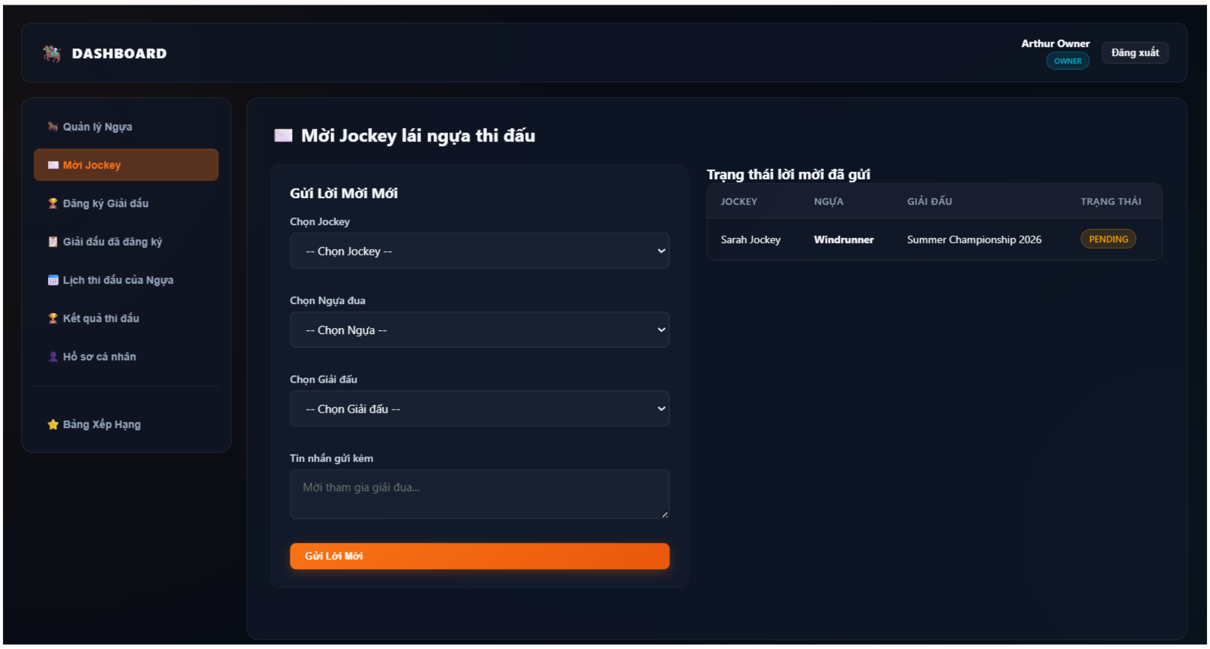

Figure 3.4: Invite Jockey

3.2.4 Register Tournament

Role: Horse Owner

- **Function trigger:** Select "Dang ky Giai dau".
- **Function description:**
  - **Purpose:**
    1. Register horse-jockey pairs.
    2. Participate in tournaments.
  - **Data processing:**
    1. Retrieve available tournaments.
    2. Check accepted jockey invitations.
    3. Select horse-jockey pair.
    4. Submit registration request.
- **Function details:**
  - Only accepted invitations may register.
  - Registration requests are sent to administrators.
  - Invalid pairs cannot participate.

Screen layout:

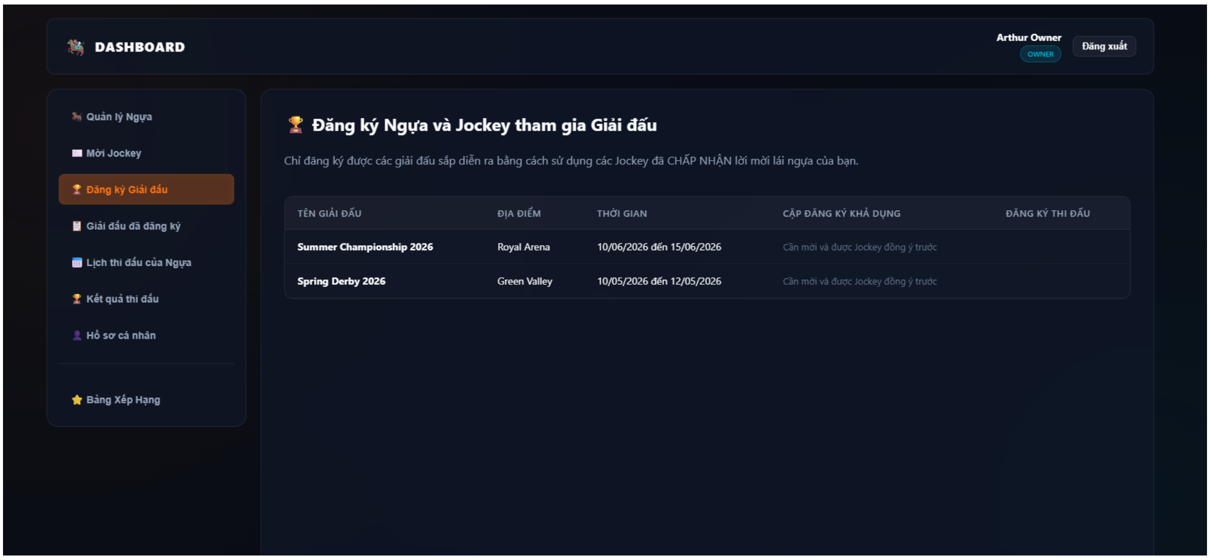

Figure 3.5: Tournament Registration

3.2.5 Registered Tournaments

Role: Horse Owner

- **Function trigger:** Select "Giai dau da dang ky".
- **Function description:**
  - **Purpose:**
    1. Monitor registration status.
    2. View approval results.
  - **Data processing:**
    1. Retrieve registration records.
    2. Retrieve approval status.
    3. Display registration history.
- **Function details:**
  - Display horse information.
  - Display jockey information.
  - Display approval status.

Screen layout:

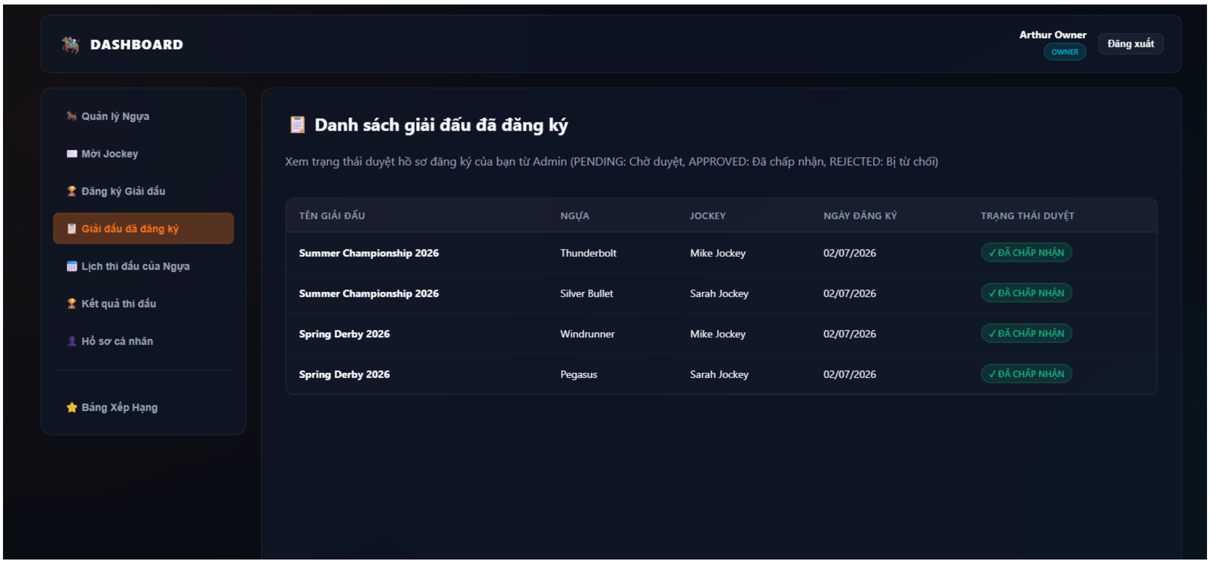

Figure 3.6: Registered Tournaments

3.2.6 Horse Schedule

Role: Horse Owner

- **Function trigger:** Select “Lich thi dau cua Ngua”.
- **Function description:**
  - **Purpose:**
    1. View upcoming races.
    2. Prepare for competitions.
  - **Data processing:**
    1. Retrieve race schedules.
    2. Retrieve tournament information.
    3. Display race information.
- **Function details:**
  - Display race date.
  - Display location.
  - Display tournament information.

Screen layout:

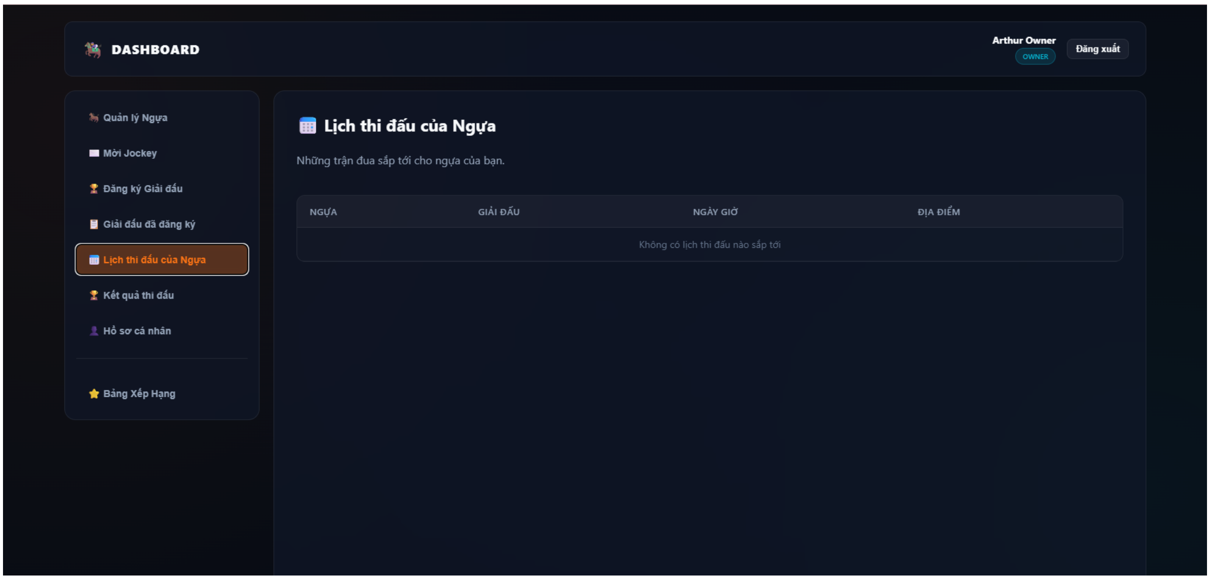

Figure 3.7: Horse Schedule

3.2.7 Race Results

Role: Horse Owner

- **Function trigger:** Select "Ket qua thi dau".
- **Function description:**
  - **Purpose:**
    1. Review race results.
    2. Evaluate horse performance.
  - **Data processing:**
    1. Retrieve completed races.
    2. Retrieve rankings.
    3. Retrieve scores.
    4. Retrieve violations.
- **Function details:**
  - Display race rankings.
  - Display accumulated points.
  - Display remarks.
  - Display violations.

Screen layout:

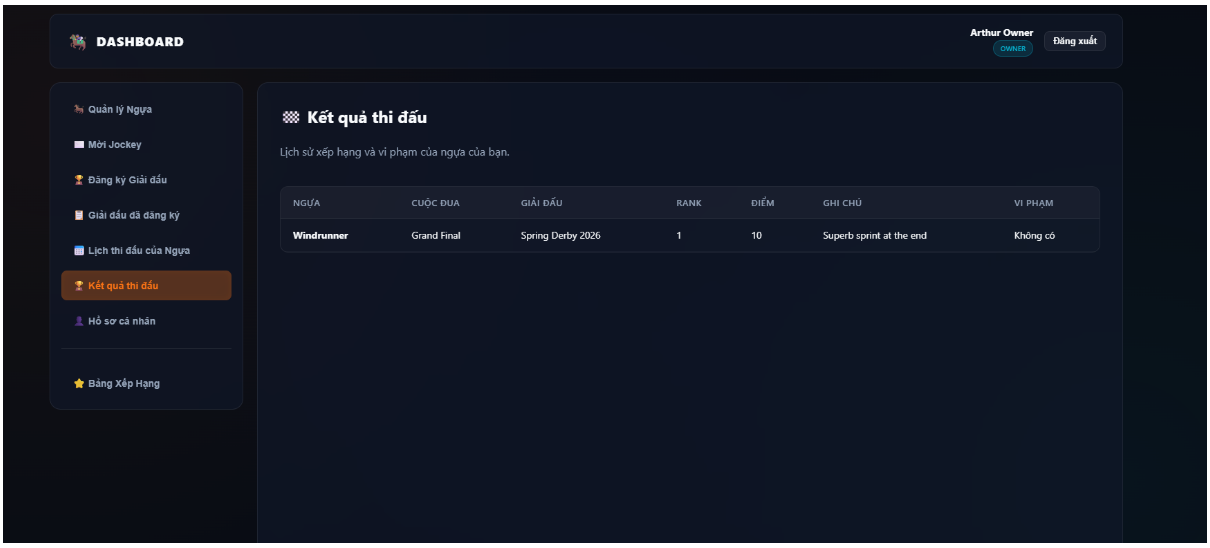

Figure 3.8: Race Results

3.2.8 Owner Profile

Role: Horse Owner

- **Function trigger:** Select "Ho so ca nhan".
- **Function description:**
  - **Purpose:**
    1. Manage personal information.
    2. Update profile information.
  - **Data processing:**
    1. Update personal information.
    2. Update contact information.
    3. Update company information.
    4. Save profile data.
- **Function details:**
  - Edit owner profile.
  - Edit biography.
  - Update social information.

Screen layout:

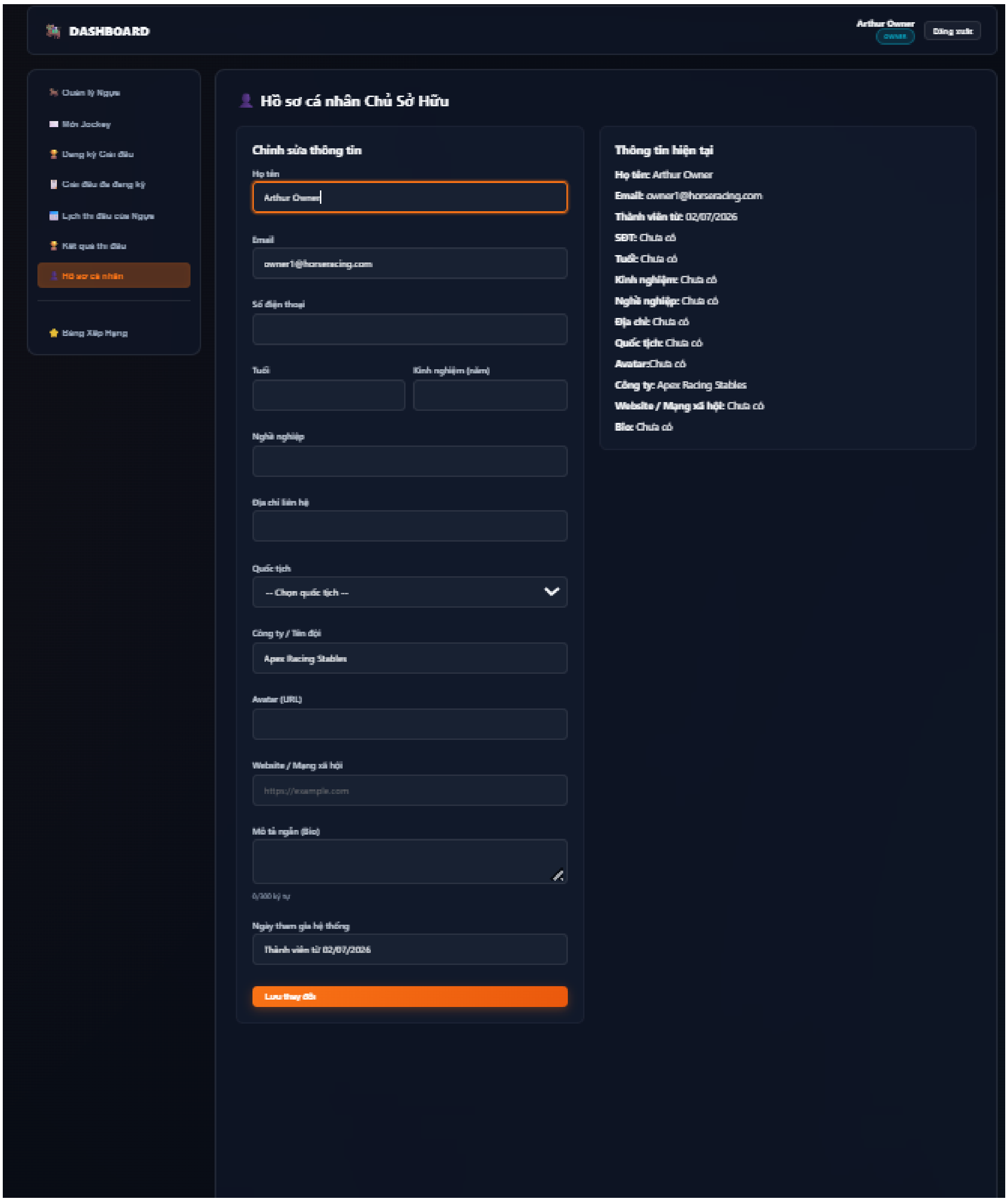

Figure 3.9: Owner Profile

3.2.9 Ranking

Role: Horse Owner

- **Function trigger:** Select "Bang xep hang".
- **Function description:**
  - **Purpose:**
    1. View official rankings.
    2. Compare competition performance.
  - **Data processing:**

1. Retrieve horse rankings.
2. Retrieve jockey rankings.
3. Filter tournaments.

- **Function details:**
  - Display ranking positions.
  - Display accumulated scores.
  - Display tournament filters.

Screen layout:

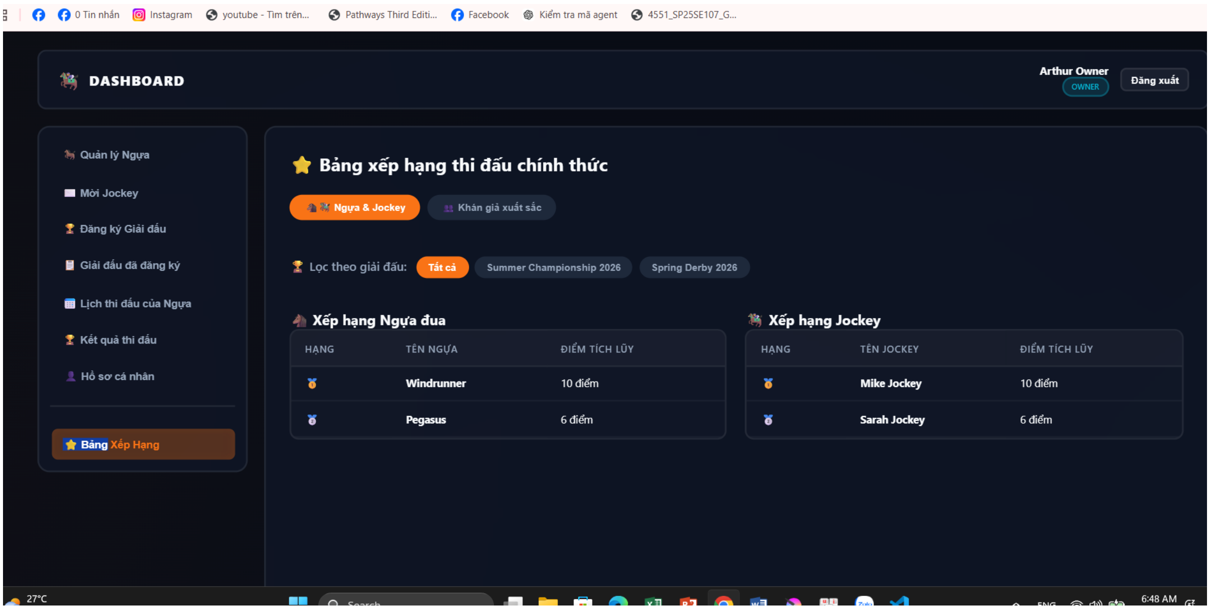

Figure 3.10: Ranking

3.3. Non-Functional Requirements

3.3.1 External Interfaces

To ensure effective communication between users and external systems, the following interfaces shall be supported.

User Interface

- The system shall provide a user-friendly and intuitive interface for Horse Owners, Jockeys, Referees, and Administrators.
- The graphical interface shall follow modern web design standards and support responsive layouts.
- Users shall be able to access the system through desktop computers, tablets, and mobile devices.
- Navigation menus and functions shall remain consistent throughout the system.

Authentication Interface

- The system shall support secure authentication using username and password.
- Role-based access control shall restrict system functions according to user roles.
- User sessions shall be managed securely to prevent unauthorized access.

Database Interface

- The application shall communicate with the database system to store and retrieve information.
- Data synchronization between the application server and database server shall be maintained.
- The database shall support CRUD operations for horses, tournaments, registrations, invitations, and users.

3.3.2 Quality Attributes

Usability

User-Friendly Interface

- The system shall provide a simple and intuitive interface.
- Functions such as tournament registration and jockey invitation shall require minimal user actions.
- The interface shall be easy to understand for both novice and experienced users.

Training and Familiarization Time

- Horse Owners and Jockeys shall require less than 30 minutes to learn the basic functions.
- Administrators shall require approximately 1–2 hours to understand management operations.

Usability Metrics

- Tournament registration shall be completed within 3 minutes.
- Error messages shall clearly guide users to correct invalid input.
- User feedback shall be collected to improve usability.

Reliability

Availability

- The system shall maintain at least 99% availability.
- Maintenance shall be scheduled during low-traffic periods.

Data Integrity

- Registration and tournament information shall be stored accurately.
- Duplicate registrations shall be prevented.
- Data consistency shall be maintained across all modules.

Error Handling

- The system shall display meaningful error messages.
- Critical system errors shall be logged for administrators.
- The system shall recover from failures with minimal data loss.

Performance

Response Time

- Normal user requests shall respond within 2 seconds.
- Under heavy load conditions, response time shall not exceed 5 seconds.

Concurrent Users

- The system shall support at least 500 concurrent users.
- Multiple users shall access the system simultaneously without significant performance degradation.

Database Performance

- Search operations shall complete within 3 seconds.
- Registration transactions shall be processed efficiently.

Security

- Passwords shall be encrypted before storage.
- Unauthorized access attempts shall be logged.
- Session timeout mechanisms shall automatically log out inactive users.
- Input validation shall prevent invalid data submission.

Maintainability

- The system shall follow a modular architecture.
- Source code shall comply with coding standards.
- New features shall be added with minimal impact on existing modules.
- API endpoints and database structures shall be documented.

Portability

- The system shall support Google Chrome, Microsoft Edge, and Mozilla Firefox.
- The application shall operate on both Windows and Linux servers.
- Users shall access the system without additional software installation.

3.4. Web Application

3.4.1 Horse Owner

Login

Role: Horse Owner

- **Function trigger:** The owner enters account credentials to log in.
- **Function description:**
  - **Purpose:**
    1. Verify the owner's identity.
    2. Grant access to owner functionalities.
  - **Data processing:**
    1. The owner enters username and password.
    2. The system validates credentials.
    3. The system identifies the OWNER role.
    4. The system creates a login session.
- **Function details:**
  - Successful login redirects to Owner Dashboard.
  - Invalid credentials display an error message.
  - Users without accounts may register.

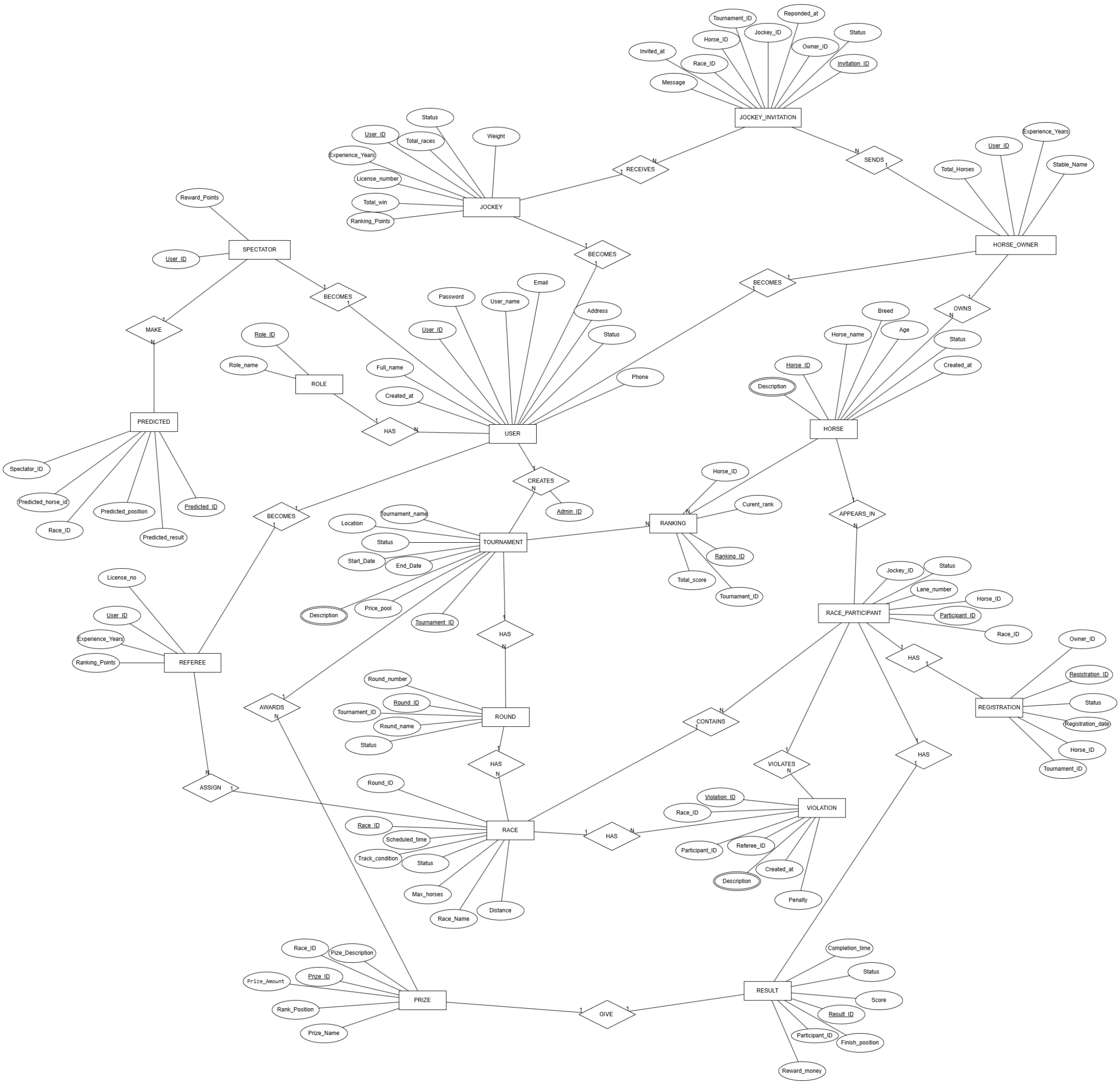

Figure 3.11: Entity Relationship Diagram

Screen layout:

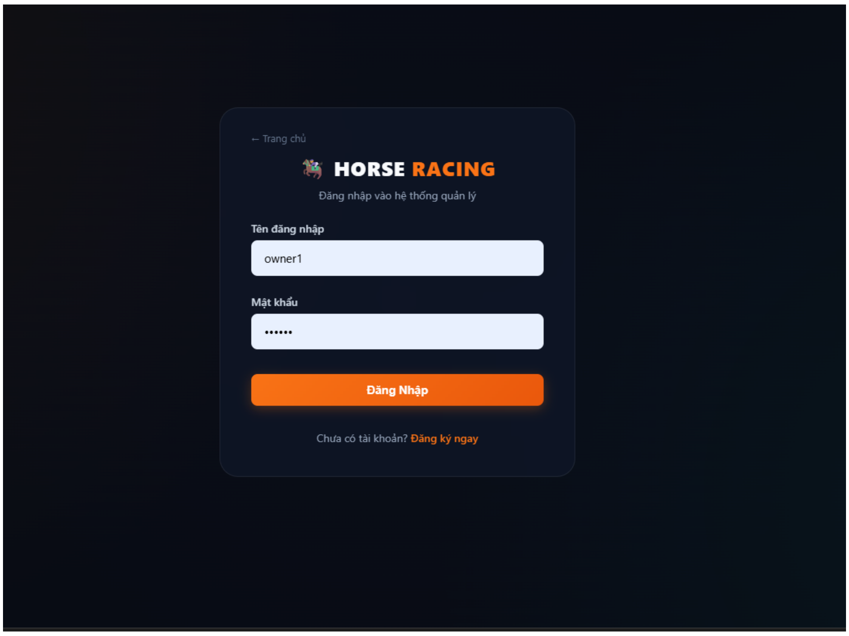

Figure 3.12: Horse Owner Login

3.4.2 Manage Horses

Role: Horse Owner

- **Function trigger:** Select "Quan ly Ngua".
- **Function description:**
  - **Purpose:**
    1. Register new horses.
    2. Manage owned horses.
  - **Data processing:**
    1. Enter horse name.
    2. Enter age.
    3. Select breed.
    4. Select gender.
    5. Save horse information.
- **Function details:**
  - Add new horses.
  - Edit horse information.
  - Delete horse information.
  - Display horse list.

Screen layout:

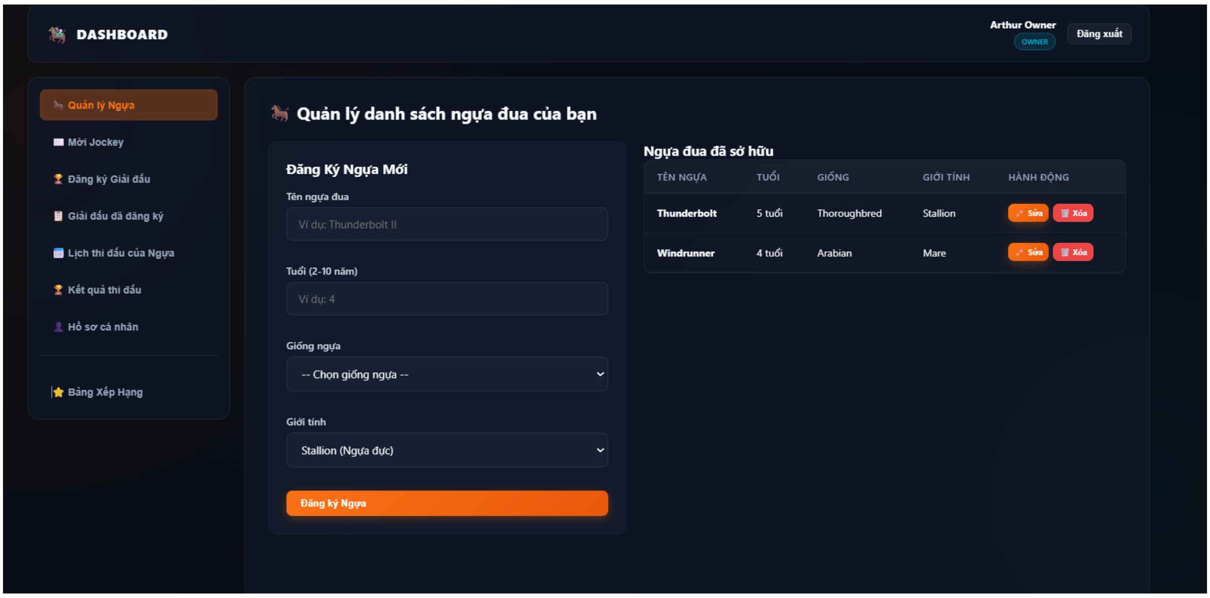

Figure 3.13: Manage Horses

3.4.3 Invite Jockey

Role: Horse Owner

- **Function trigger:** Select "Moi Jockey".
- **Function description:**
  - **Purpose:**
    1. Invite jockeys.
    2. Create horse-jockey partnerships.
  - **Data processing:**
    1. Select jockey.
    2. Select horse.
    3. Select tournament.
    4. Enter invitation message.
    5. Send invitation.
- **Function details:**
  - Invitation status is displayed.
  - Pending invitations can be monitored.
  - Accepted invitations may be used for registration.

Screen layout:

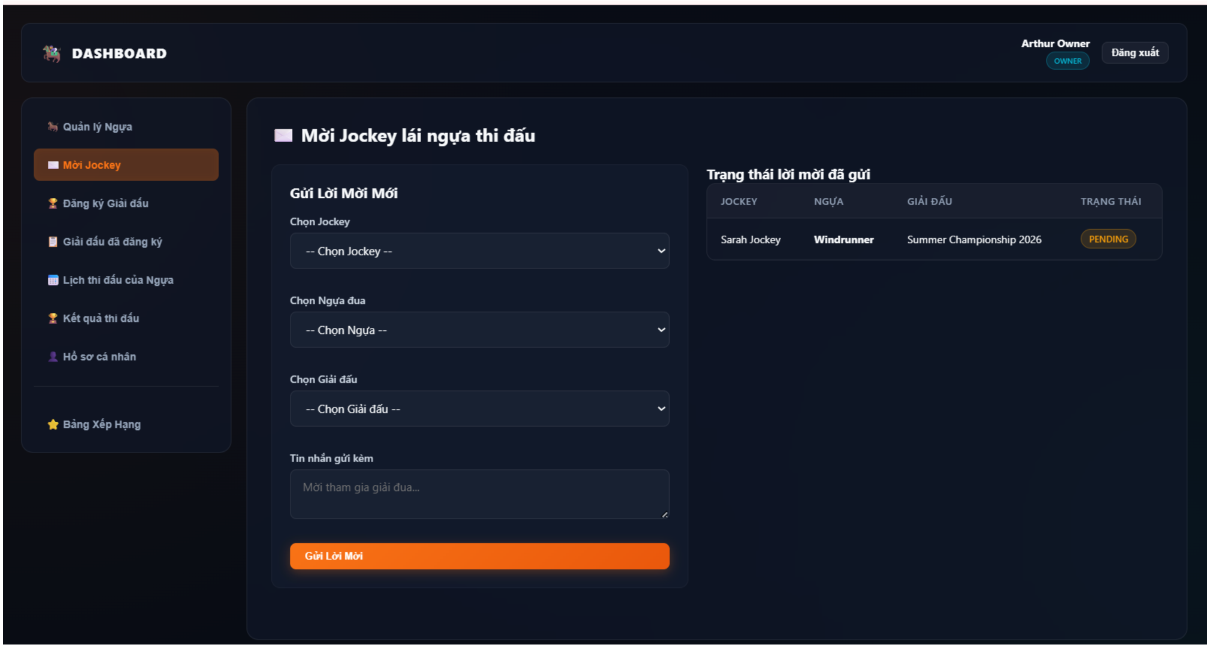

Figure 3.14: Invite Jockey

3.4.4 Register Tournament

Role: Horse Owner

- **Function trigger:** Select "Dang ky Giai dau".
- **Function description:**
  - **Purpose:**
    1. Register horse-jockey pairs.
    2. Participate in tournaments.
  - **Data processing:**
    1. Retrieve available tournaments.
    2. Check accepted jockey invitations.
    3. Select horse-jockey pair.
    4. Submit registration request.
- **Function details:**
  - Only accepted invitations may register.
  - Registration requests are sent to administrators.
  - Invalid pairs cannot participate.

Screen layout:

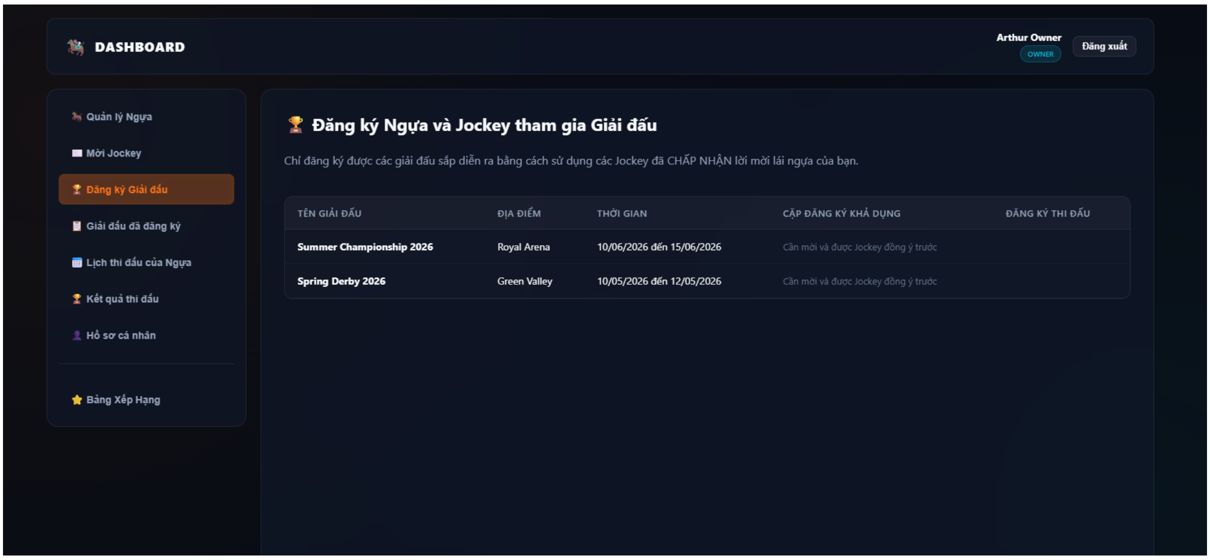

Figure 3.15: Tournament Registration

3.4.5 Registered Tournaments

Role: Horse Owner

- **Function trigger:** Select "Giai dau da dang ky".
- **Function description:**
  - **Purpose:**
    1. Monitor registration status.
    2. View approval results.
  - **Data processing:**
    1. Retrieve registration records.
    2. Retrieve approval status.
    3. Display registration history.
- **Function details:**
  - Display horse information.
  - Display jockey information.
  - Display approval status.

Screen layout:

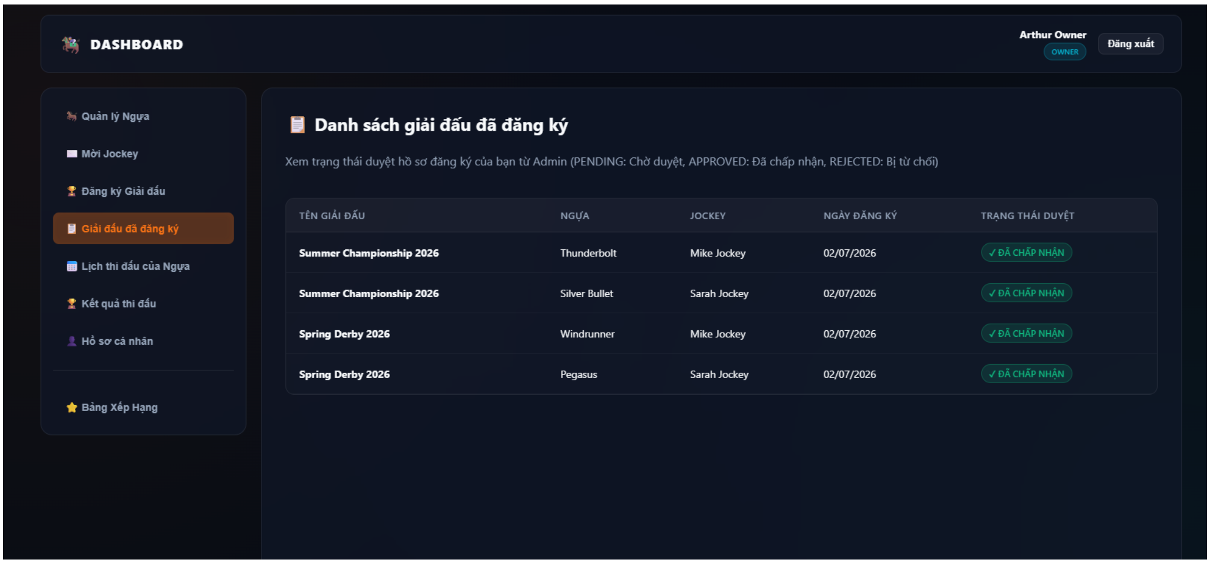

Figure 3.16: Registered Tournaments

3.4.6 Horse Schedule

Role: Horse Owner

- **Function trigger:** Select “Lich thi dau cua Ngua”.
- **Function description:**
  - **Purpose:**
    1. View upcoming races.
    2. Prepare for competitions.
  - **Data processing:**
    1. Retrieve race schedules.
    2. Retrieve tournament information.
    3. Display race information.
- **Function details:**
  - Display race date.
  - Display location.
  - Display tournament information.

Screen layout:

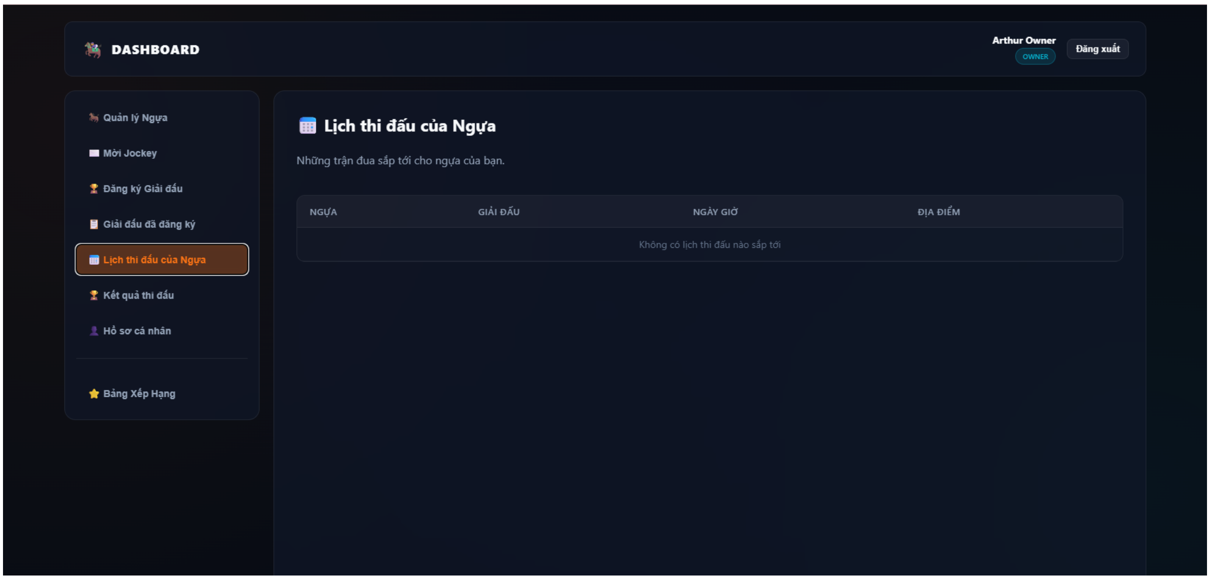

Figure 3.17: Horse Schedule

3.4.7 Race Results

Role: Horse Owner

- **Function trigger:** Select "Ket qua thi dau".
- **Function description:**
  - **Purpose:**
    1. Review race results.
    2. Evaluate horse performance.
  - **Data processing:**
    1. Retrieve completed races.
    2. Retrieve rankings.
    3. Retrieve scores.
    4. Retrieve violations.
- **Function details:**
  - Display race rankings.
  - Display accumulated points.
  - Display remarks.
  - Display violations.

Screen layout:

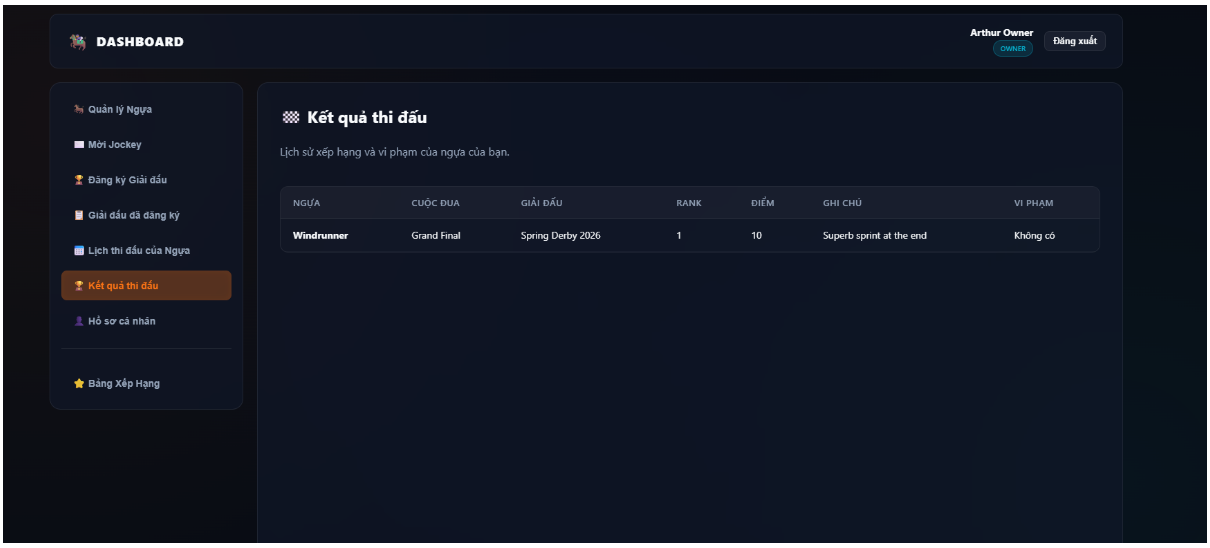

Figure 3.18: Race Results

3.4.8 Owner Profile

Role: Horse Owner

- **Function trigger:** Select "Ho so ca nhan".
- **Function description:**
  - **Purpose:**
    1. Manage personal information.
    2. Update profile information.
  - **Data processing:**
    1. Update personal information.
    2. Update contact information.
    3. Update company information.
    4. Save profile data.
- **Function details:**
  - Edit owner profile.
  - Edit biography.
  - Update social information.

Screen layout:

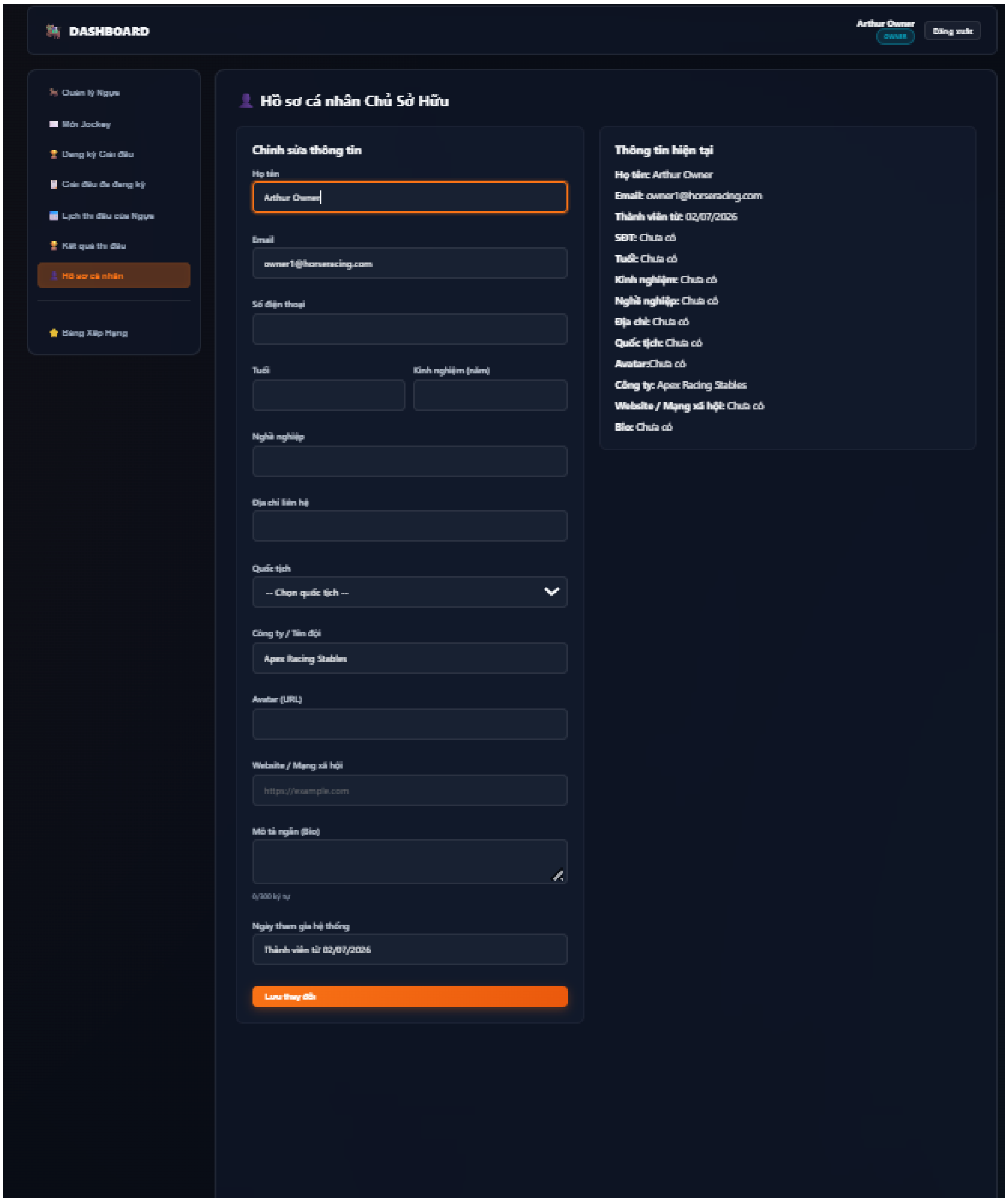

Figure 3.19: Owner Profile

3.4.9 Ranking

Role: Horse Owner

- **Function trigger:** Select “Bang xep hang”.
- **Function description:**
  - **Purpose:**
    1. View official rankings.
    2. Compare competition performance.
  - **Data processing:**

1. Retrieve horse rankings.
2. Retrieve jockey rankings.
3. Filter tournaments.

- **Function details:**
  - Display ranking positions.
  - Display accumulated scores.
  - Display tournament filters.

Screen layout:

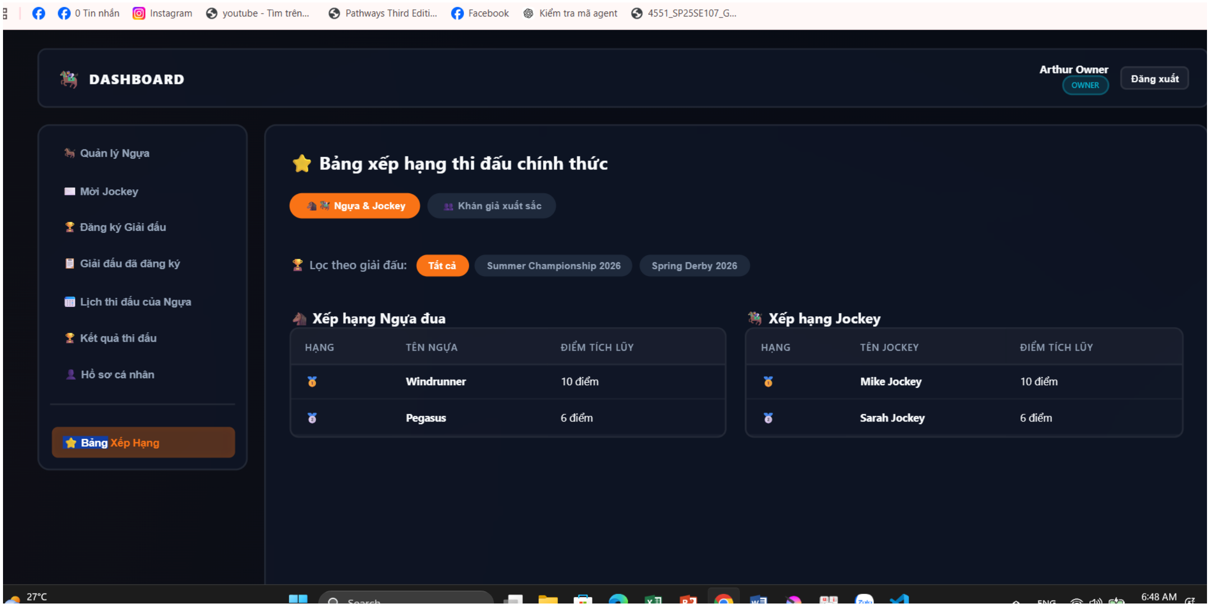

Figure 3.20: Ranking

IV. Software Design Description

This chapter presents the software architecture, database design and detailed design.